

WINGS AIRLINE
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Passengers' Satisfaction Report

SHANGHAI - TEAM J

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Business Background & Mission Statement

Wings Airline is an **airline business that has only been operating for 1 year**, They have a plan to develop the business by collaborating with several types of travel agents and develop more features, Mr. Haris as the **Director of Services and Commercial of Wings Airline**, asked the data analyst team to:

- **Make analysis of the airline's touch point factor performance to passengers during the business.**
- **Identify the most significant touch point factor that needs to be improved in order to increase customer loyalty to Wings Airline by improving customer satisfaction by 25% within 12 months.**
- **Develop a cluster analysis to know who is our target market, so the management could select the best additional feature for our targeted market.**



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Why Passengers Satisfaction is so important?

- Leads to loyalty
- Maintain a positive brand image
- Attracts new customers

Source: Ahmed, Rizwan Raheem; Vveinhardt, Jolita; Warraich, Usman Ali; Hasan, Syed Shabib ul; Baloch, Akhtar. Customer Satisfaction & Loyalty and Organizational Complaint Handling: Economic Aspects of Business Operation of Airline Industry. *Inzinerine Ekonomika-Engineering Economics, 2020, 31 (1).*

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Data Set Overview



Data Scope:

Survey data on passengers satisfaction are from **130,000 passenger** records, which contains detailed passenger data and their satisfaction ratings.

129.880
Passenger

- 51% Female, 49% Male
- 82% Loyal Customer
- 69% Business Travel
- The Passenger Age Range: between 7 - 85 Years Old

4
Products

- Eco Class
- Eco Plus Class
- Business Class
- Flight Distance range 31 Km - 4.983 Km

24
Columns

- 15 Questionnaire Data
- 4 Passenger Data
- 5 Airline Data

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Data Process

Define Objectives

Determine the objective, scope, and root cause

Data Gathering

Conduct data gathering and information about Airline bussiness

Data Preparation

Conduct data cleaning using Python

Data Analysis

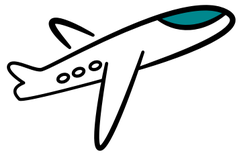
Conduct Data Analysis using Phyton and Tableau

Recommendations

Recommend the suitable recommendations

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Satisfaction



Service

Pre-flight

- Online
  - Online Booking
  - On-board Service
  - Online Boarding
- Offline
  - Gate Location
  - Check-in service

Inflight

- Inflight wifi service
- Food and drink
- Seat comfort
- Inflight entertainment
- Leg room service
- Inflight service
- Cleanliness

Post-flight

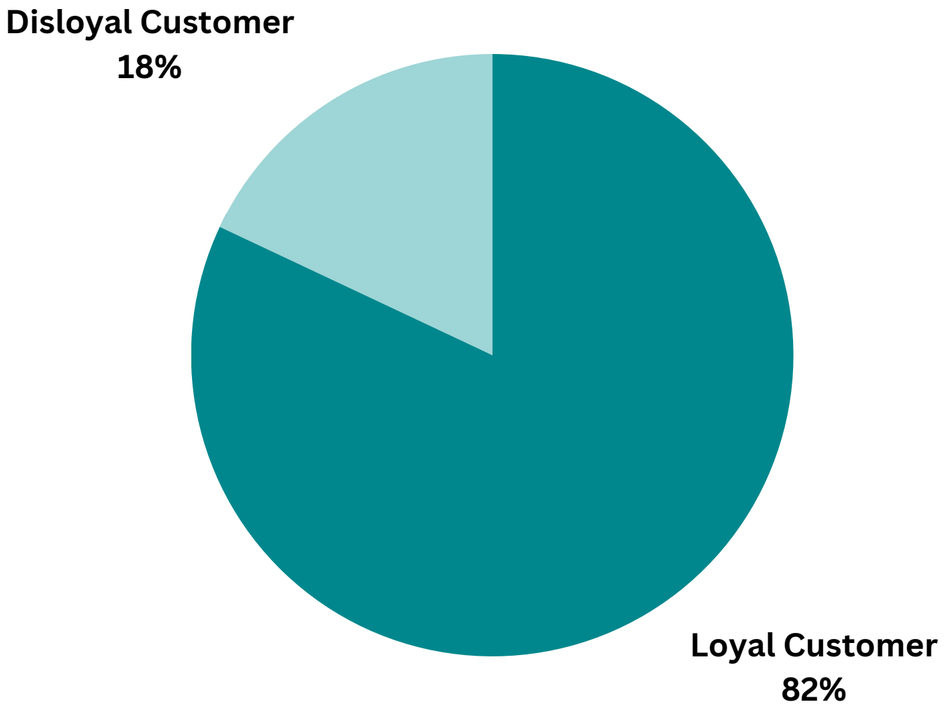
- Baggage handling
- Departure/Arrival time convenient
- Departure Delay in Minutes
- Arrival Delay in Minutes

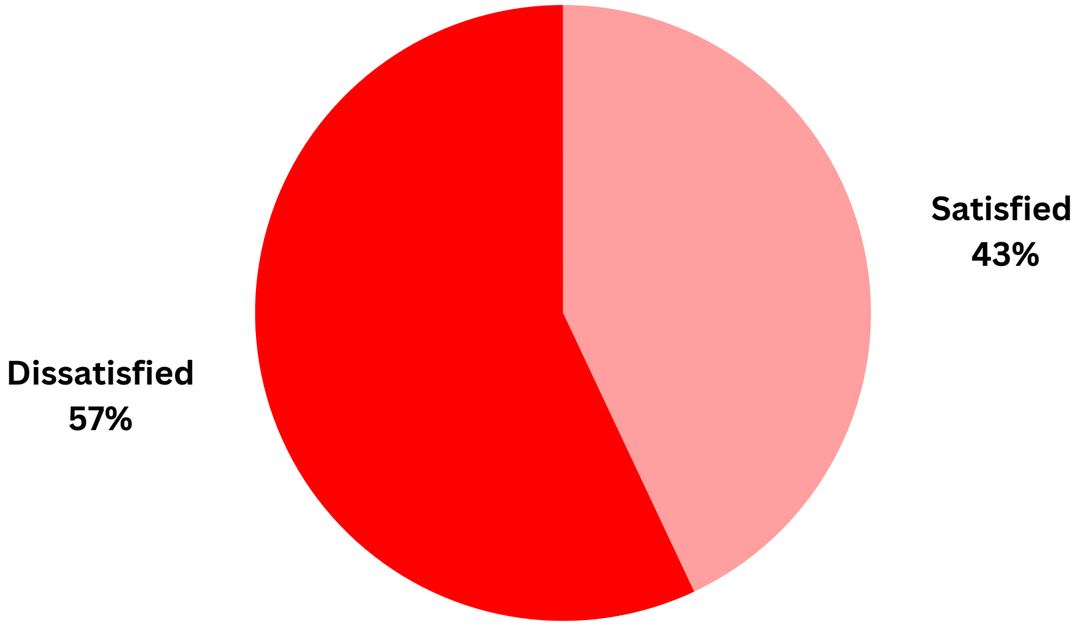
Product

- Distance
- Class

Current Situation

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Executive Summary

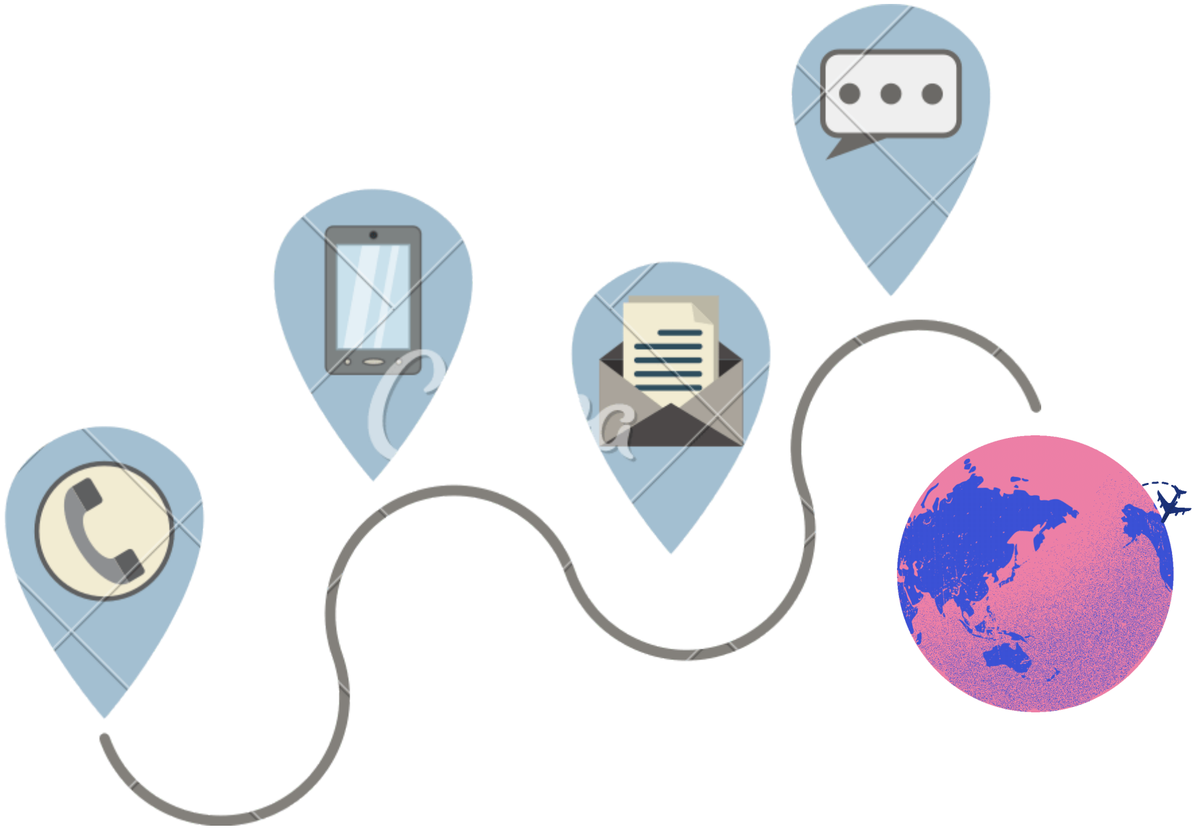
- Low (43%) passengers satisfaction
- Lowest passenger satisfaction score is in **Economy Class**
- Majority of dissatisfied passengers are on **short-haul flights**

Key touch point that need to be enhanced:

- Pre-flight: Online services & Gate Location
- Inflight: Cleanliness, Inflight Wi-fi Service, Food and Drink service.
- Post-flight: Departure and Arrival Time Convenient

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Key Touch Point Analysis



Consumer Journey

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Pre-Flight



Online Service



Offline Service

Inflight



Inflight Service

Post-flight



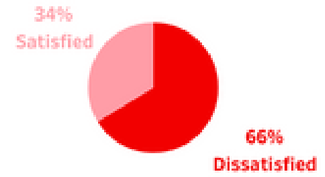
Post-flight Service



Online Service Satisfaction

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Ease_of_Online_booking



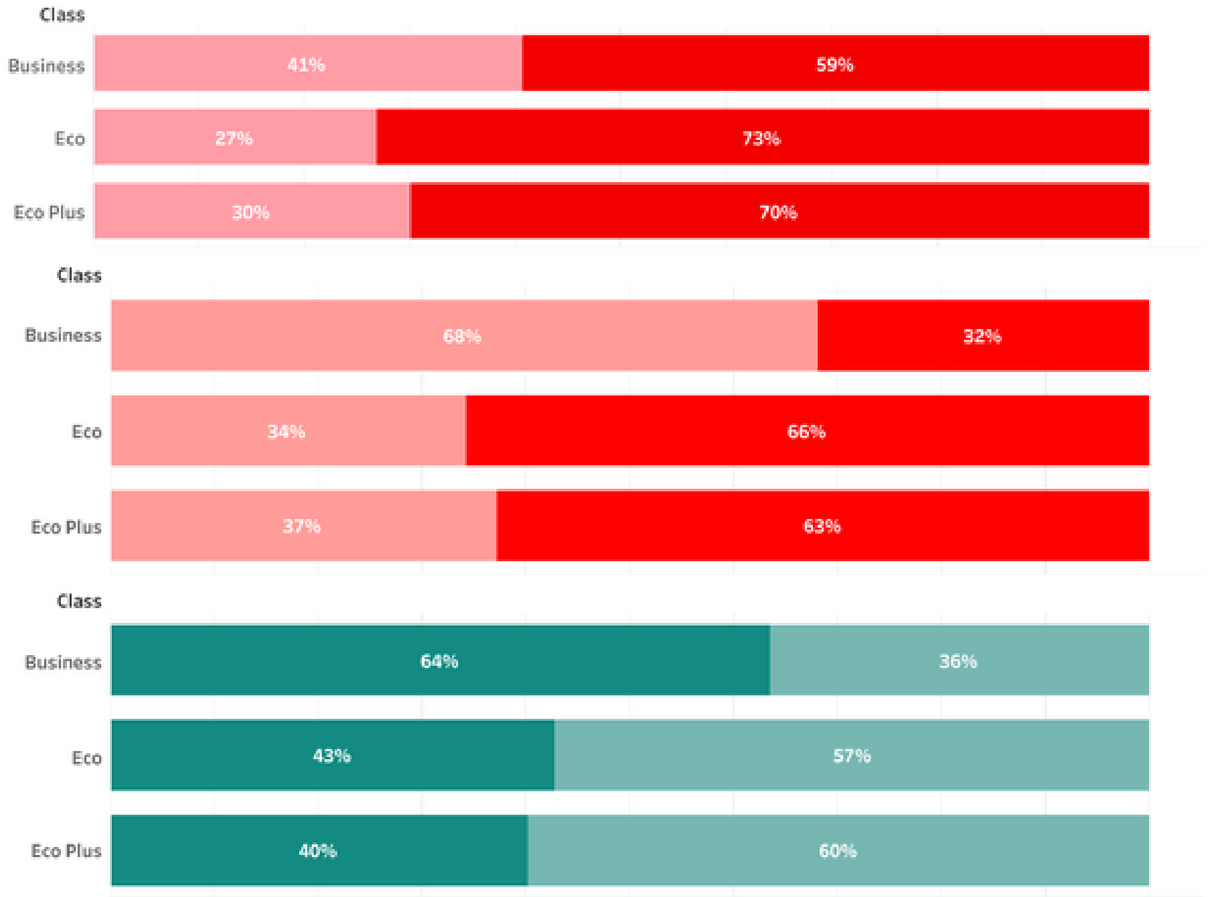
Online Boarding



On-board_service



Ease of online booking(class)

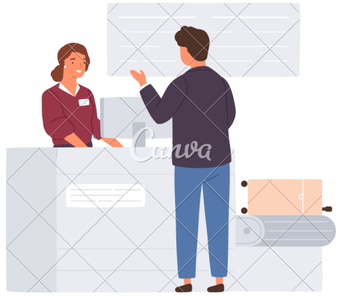


Insight:

- **Economy and Economy Plus Class** have the most dissatisfied passenger with our **Online Booking & Online boarding service**

Possible cause:

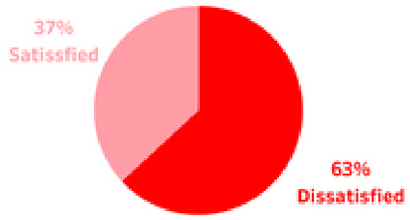
- Slow loading page
- Unloaded page
- Lack of payment option
- Lack of Promotion

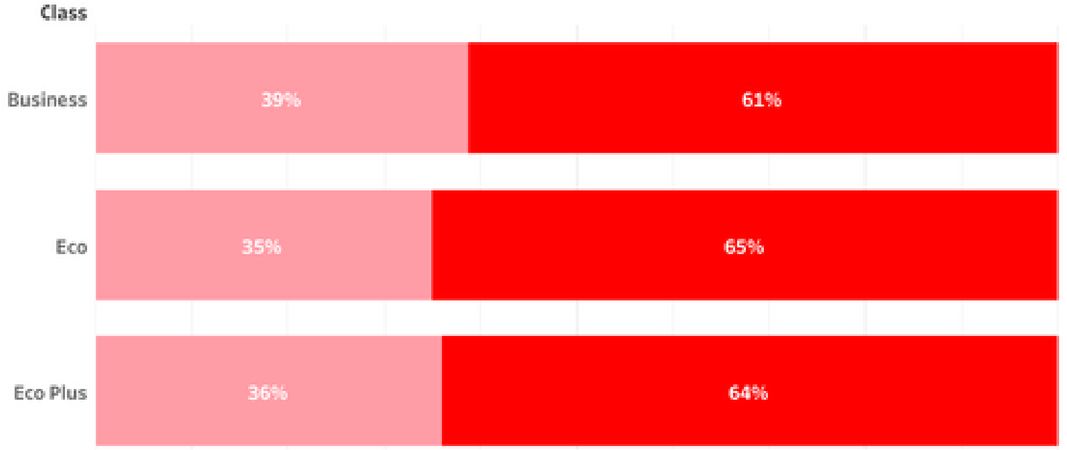


Offline Service Satisfaction

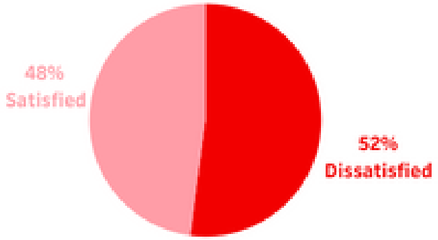
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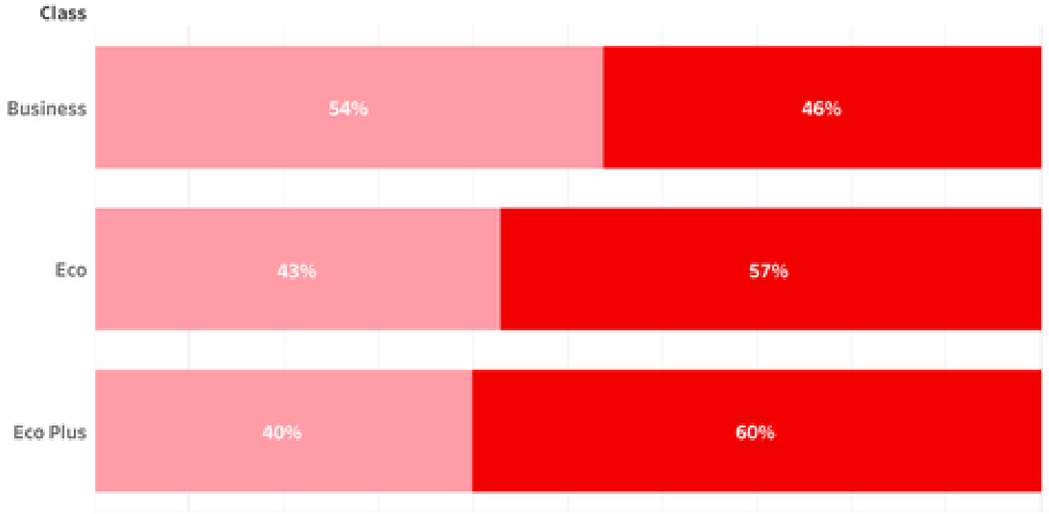
Gate_location





Checkin Service





Insight:

- Passengers from every class are not happy with our **Gate Location** & Passengers from the Economy and Economy Plus class are not happy with our **Check-In service**

Possible cause:

- Unfriendliness and Slow service
- Far Gate Location



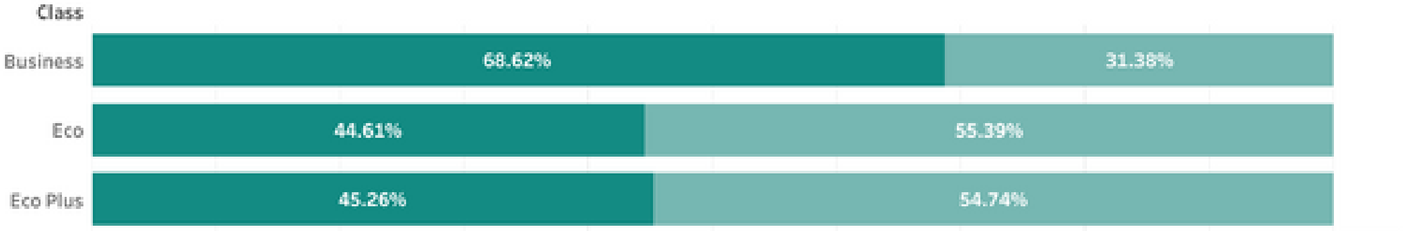
Inflight Service Satisfaction

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Seat_comfort



Seat_comfort(Class)



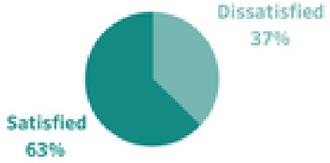
Inflight entertainment



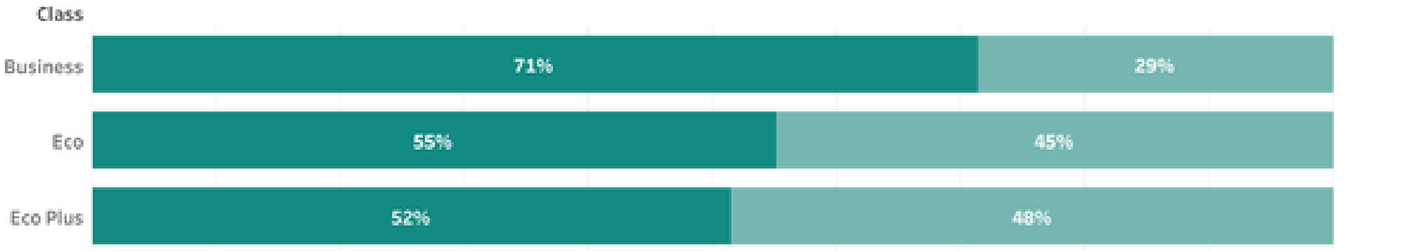
Inflight entertainment(Class)



Inflight_service



Inflight Service(Class)



Insight:

- More than half passengers of Economy and Economy Plus class are still dissatisfied with their Seat Comfort and Inflight Entertainment



Inflight Service Satisfaction

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Food_and_drink



Food n Drink(Class)



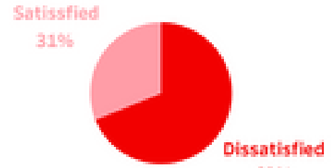
Cleanliness



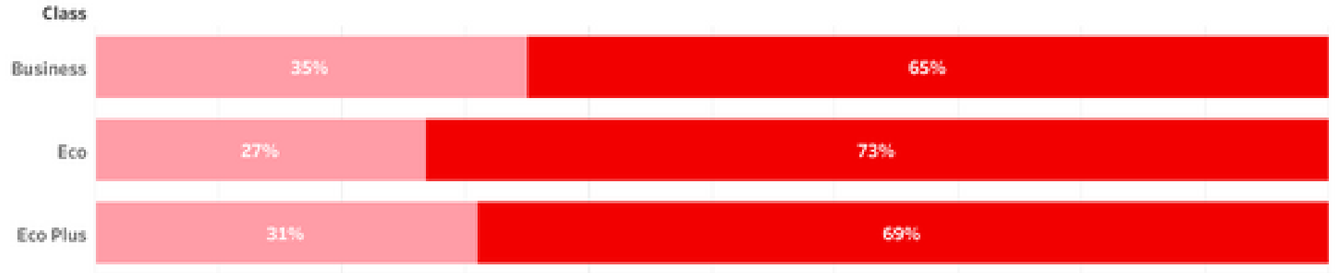
Cleanliness (Class)



Inflight Wifi Service



Inflight Wifi Service(Class)

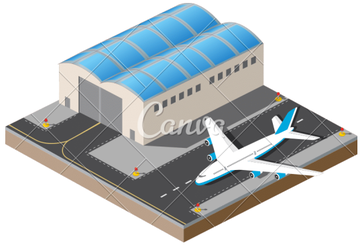


Insight:

- Economy and Economy Plus class passengers are not happy with the **Food and Drink service, Inflight Wi-Fi service, and Cleanliness.**
- Business Class passengers are dissatisfied with **Inflight Wi-Fi service**

Possible cause:

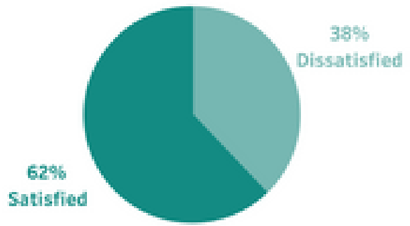
- Dirty seat and folded table
- Business Class:
  - Slow connection
  - Narrow variety of food & drink option
- Economy Class:
  - Unavailable Wi-fi service
  - Only provided with cup of mineral water

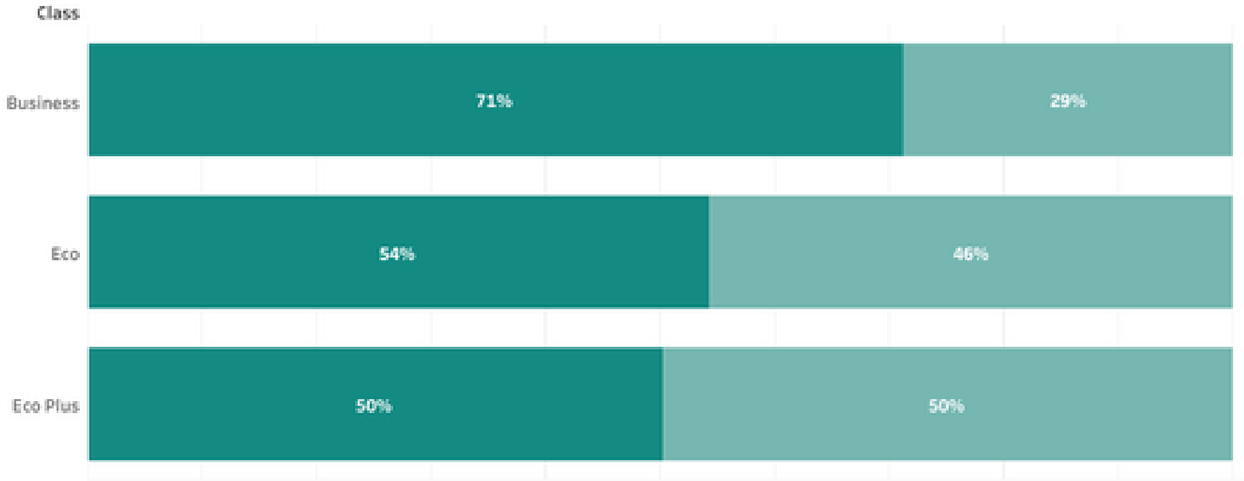


Post-flight Service Satisfaction

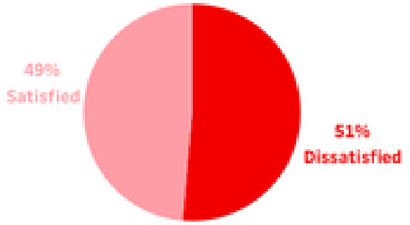
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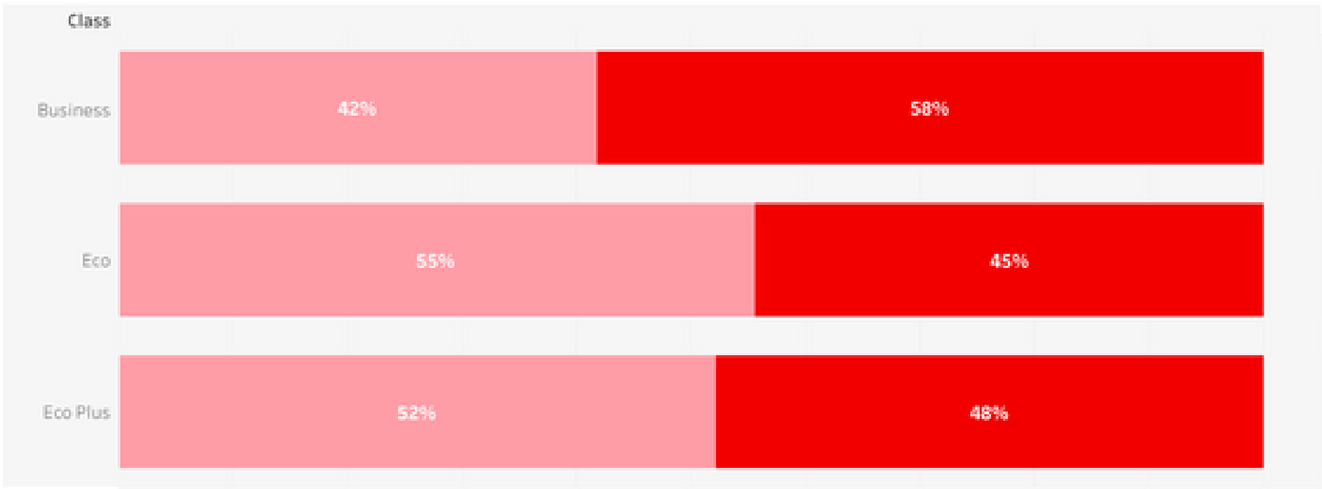
Bagage Handling





Departure/Arrival_time_convenient



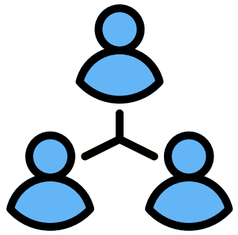


Insight:

- Business class passengers are not happy with the **Time Convenient** of our flight

Possible cause:

- Late departure and arrival would lead to late meetings



Variable that has bigger impact with satisfaction

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Statistical Analysis is conducted to determine the priority of the service that has the bigger impact on satisfaction and need to be improved. the analisys are:

- **Chi-square Analysis**,
- **Correlation Analysis**
- **p-value**

we also make models using **machine learning** in Phyton for predictive analysis with **Logistic Regression, RandomForest, dan XGBoost** and the model that have the highest Accuracy is from random forest which is 0.9560326408731691 / 95.6%

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Conclusion



Touch points that needs to be fixed

- Online Service for Economy and Economy Plus Class
- Cleanliness of our board for Economy & Economy Plus Class
- Inflight Wi-fi Service in every Class
- Check-In Process for Economy & Economy Plus Class
- Food & Drink Service in every Class
- Our Gate Location for every Class
- The Arrival and Departure convenient for Business Class

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Recommendation

Pre-Flight

- Partnership with OTA (Online Travel Agent) to reduce server maintenance costs and added more value to our online service

traveloka pegipegi

- Provide golf cart support for our passengers to reach their gate locations



- Implementing guest service procedures and best practices for maintaining a friendly and quick service for our passenger

Inflight

- Inflight Wi-Fi Service:
  - **Business Class**: Changing internet service provider for better connection
  - **Eco & Eco+ Class**: Offer Wi-Fi service with reasonable pricing
- Providing a variety of content, such as free music, free games, and movies rent with reasonable pricing
- Offer vaster variety of food and drink options, including vegetarian and vegan options.
- Implementing cleaning procedures and best practices for maintaining a clean and safe cabin by our flight attendants and other crew members

Post- flight

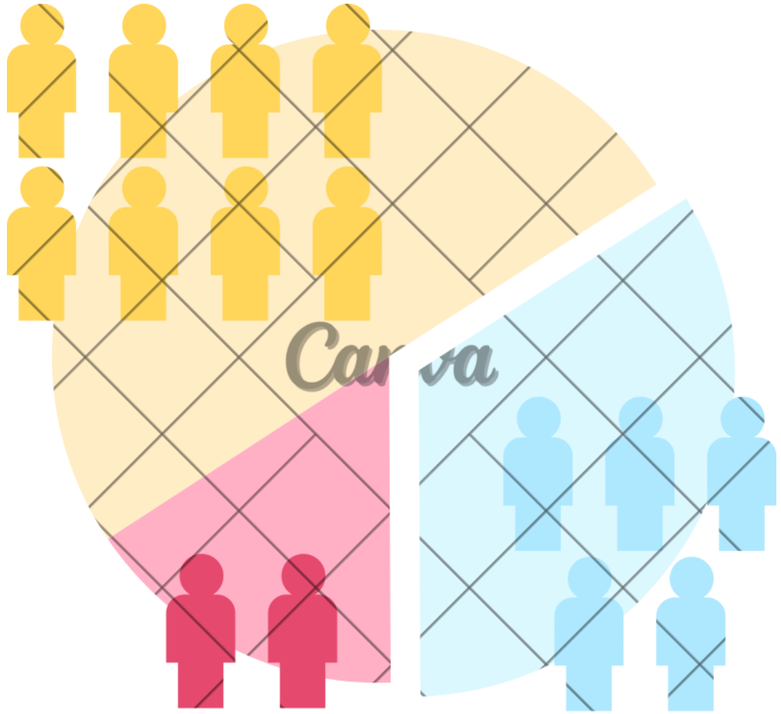
- Minimize delays



- Improve communication



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Customer Segmentation & Target Market

Customer Segmentation

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Traveling Families

- Age Between 7 - 70
- Eco Class
- Very Short - Med Haul Travel
- Dissatisfied



Senior Professional

- Age Between 35 - 40
- Business or Economy Class
- Very Long Haul Travel
- Dissatisfied



The Executives

- Age Between 40 - 60
- Business Class
- Short - Long Haul Travel
- Satisfied



Young Professional

- Age Between 22 - 25
- Use Eco/Eco Plus Class
- Short Haul Travel
- Dissatisfied

Target Market

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Traveling Families

- Age Between 7 – 70
- Eco Class
- Very Short – Med Haul Travel
- Dissatisfied



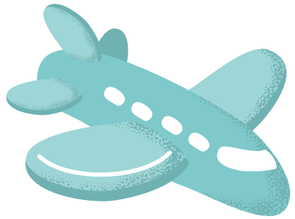
The Executives

- Age Between 40 – 60
- Business Class
- Short – Long Haul Travel
- Satisfied

Based on the Pareto Chart, the ‘Traveling Families’ and ‘The Executives’ will be our priority customer

For our marketing campaign we are going to prioritize the ‘Senior Level’ and ‘Junior Level’ cluster

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Flight Planner



The Executives

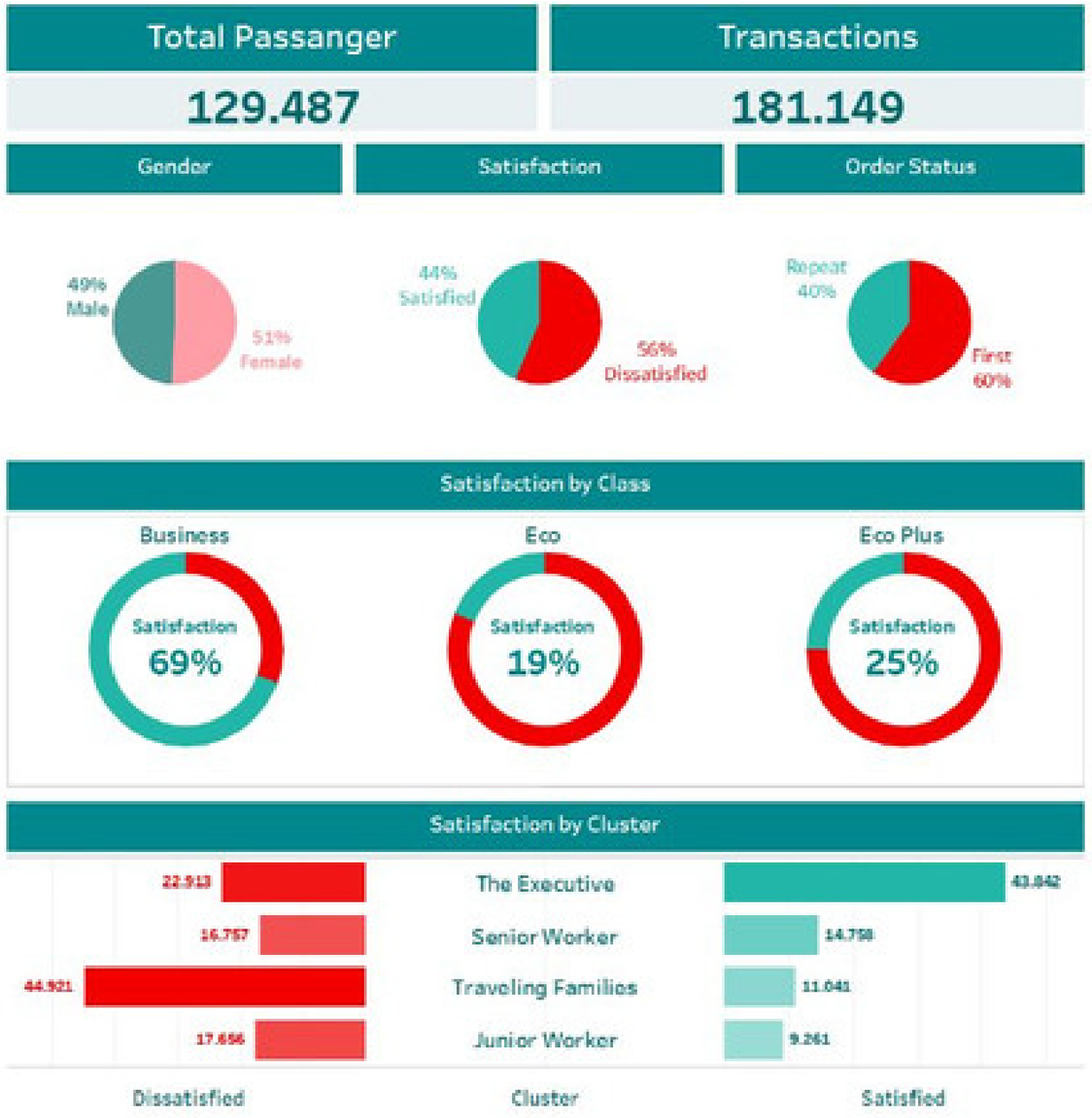
- Executive Lounge
- Assisted Check-In and Boarding
- Pre-book Meal
- Baggage Delivery

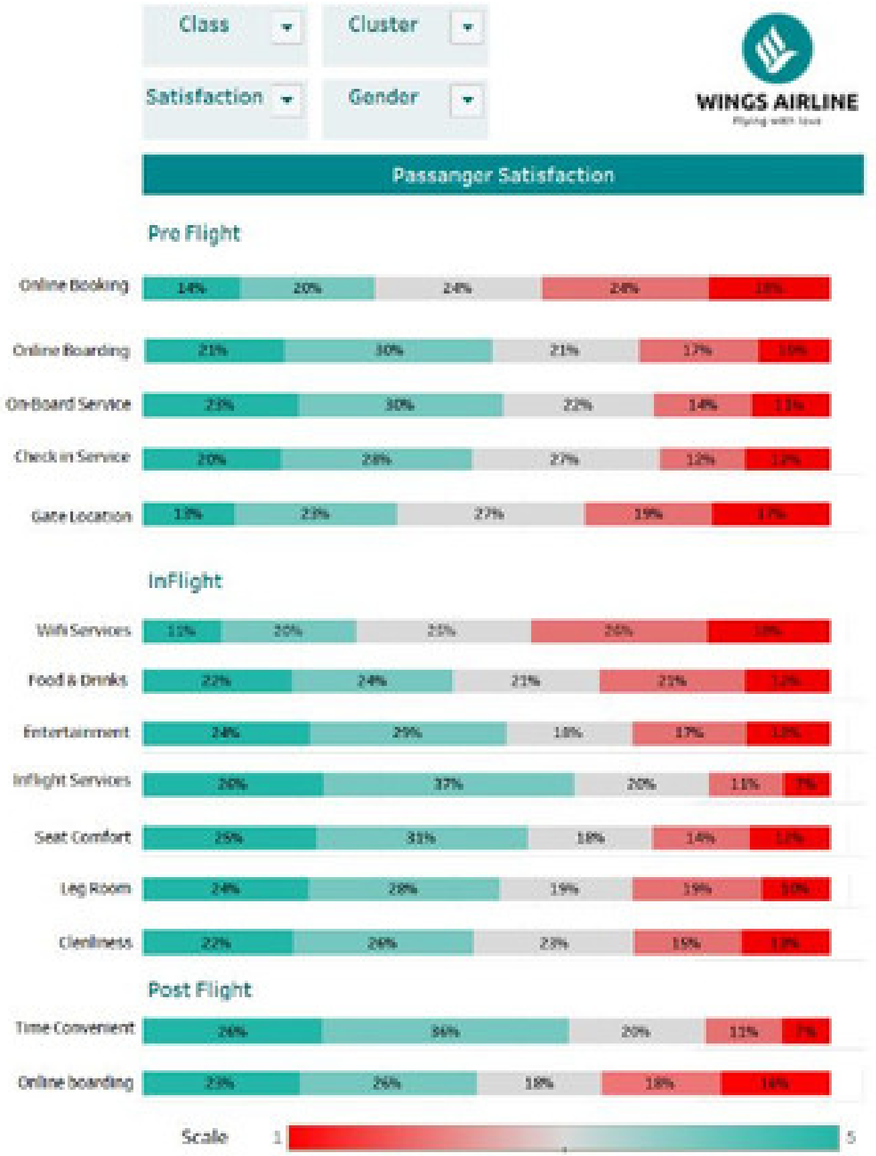


Traveling Families

- Additional Baggage Allowance
- Early boarding for families
- Special meals for children
- In-flight entertainment for kids

DASHBOARD


Total Passanger
129.487
Transactions
181.149
Gender
Satisfaction
Order Status
49% Male
51% Female
44% Satisfied
56% Dissatisfied
Repeat 40%
First 60%
Satisfaction by Class
Business
Satisfaction 69%
Eco
Satisfaction 19%
Eco Plus
Satisfaction 25%
Satisfaction by Cluster
22.913
16.757
44.921
17.656
The Executive
Senior Worker
Traveling Families
Junior Worker
43.842
14.758
11.041
9.261
Dissatisfied
Cluster
Satisfied


Class
Cluster
Satisfaction
Gender
WINGS AIRLINE
Passanger Satisfaction
Pre Flight
Online Booking
Online Boarding
OnBoard Service
Check in Service
Gate Location
InFlight
Wifi Services
Food & Drinks
Entertainment
Inflight Services
Seat Comfort
Leg Room
Cleanliness
Post Flight
Time Convenient
Online boarding
Scale
1
5

Recommendation for Future Analysis

- Data from accounting to gain analysis and insight on Ticket Pricing and Revenue
- Data from IT team to gain analysis and insight on how many time does each passenger go to our website and and how long did it takes to book a ticket for each passenger

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THANKYOU FOR YOUR ATTENTION!



Flying with love and safely with Wings Airline!

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APPENDIX